MILITARY SPECIFICATION

CASE, DENTAL INSTRUMENT AND SUPPLY SET, EMPTY

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers an aluminum container for instruments necessary to afford emergency dental treatment under field conditions.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-A-318 — Aluminum Alloy Plate and Sheet 5052.

TT-T-306 — Thinner; Synthetic-Enamel.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Box, Fiberboard.

PPP-B-676 — Boxes, Set-Up, Paperboard.

TT-E-527 — Enamel, Alkyd, Lustreless.

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MIL-C-5541 — Chemical Films for Aluminum and Aluminum Alloys.

MIL-P-8585 — Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

MIL-A-8625 — Anodic Coatings, for Aluminum and Aluminum Alloys.

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper, Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 — Metals; Test Methods.

Fed. Std. No. 595 — Colors.

FSC 6545

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MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MS16686 — Medical Services Symbol.

DRAWING

MILITARY MEDICAL SUPPLY AGENCY

1848 — Case, Dental Instrument, and Supply Set, Empty.

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Component parts.** The case consists of a body, a cover, and a removable center plate. Within the body are welded several partitions and instrument retainers, called "Body Inserts." To the center plate are welded instrument supports, called "Center Plate Inserts."

3.2 **Materials.** Except as herein specified, materials shall conform to the applicable portions of the specifications, standards, and drawing listed in section 2, and shall be capable of meeting the requirements shown therein. In addition, the metal shall, conform to the chemical requirements of their applicable specifications when tested as specified in 4.2.3. When materials to be used are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended.

3.2.1 *Aluminum alloy sheet.* Aluminum alloy sheet shall conform to Specification QQ–A–318.

3.2.2 *Primer.* Primer shall conform to Specification MIL–P–8585, and shall be color Y.

3.2.3 *Enamel.* Enamel shall conform to Specification TT–E–527. Color shall be green, color 34087 as shown in Standard 595, except that for marking, as specified in 3.6, color shall be black, color 37038 of the same standard (see 6.2).

3.3 **Construction.** Construction shall conform to the requirements specified herein and to those shown in Drawing 1848.

3.3.1 *Construction details.* Case shall be fabricated from aluminum alloy sheet as specified in 3.2.1, using temper O for the body and cover, and temper H22 or H32 for center plate, body inserts, and center plate inserts. Body and cover shall each be die-drawn and trimmed so as to be seamless, and one-piece.

3.4 **Welding.** Welding shall be done by qualified welders using appropriate equipment in a qualified process. Maximum distance between spot weld centers shall be 2 inches where two or more spot welds are used to attach one piece to another. Spot welds shall be capable of developing the tensile strength of the parent metal. All spot welds shall be cleaned and ground to blend with adjacent surfaces.

3.5 **Finish.** All accessible aluminum alloy surfaces of the case, both inside and outside, shall be finished with enamel as specified herein. Surface to be enameled shall be thoroughly cleaned and chemically degreased, then prepared for topcoating by one of the following three methods:

(1) Chemical treatment in accordance with Specification MIL–C–5541.

(2) Anodic treatment in accordance with Specification MIL–A–8625.

(3) Priming treatment, using material specified in 3.2.2, thinned and

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applied by spray in accordance with the directions specified under "Preparation for Delivery" in Specification MIL–P–8585, and air-dried for not less than 30 minutes before topcoating.

Topcoating shall be with one coat of enamel as specified in 3.2.3. The enamel shall be reduced by taking four parts by volume of enamel with one part by volume of thinner in conformance with Specification TT–T–306. The reduced enamel shall be applied by spraying and cured by a heat treatment by baking at 250° Fahrenheit for not less than 45 minutes. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, or cracking.

Contractor shall endeavor to control film thickness so that the thickness of a single dried coat is 0.25 to 0.35 mil for primer (if used) and 1.0 to 1.5 mil for enamel.

3.6 Marking for identification. Each case shall be permanently and legibly marked with enamel as specified in 3.2.3, as follows:

Centrally located on the outside surface of the cover, a 1-inch diameter Medical Services Symbol as shown for type I in Standard MS16686.

Below the Medical Services Symbol, in vertical letters, ¼-inch high, the legend: "DENTAL INSTRUMENTS".

On the inside of the cover, in vertical letters, ¼-inch high, the name or registered trade-mark of the manufacturer, and the contract number or order number.

3.7 Workmanship. Workmanship shall be first class throughout. Cases shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component, or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 *For examination.* Sampling for examination shall be conducted in accordance with Standard MIL–STD–105.

4.2.2 *For tests.* Sampling for tests shall be conducted in accordance with Standard MIL–STD–105.

	Inspection level	*AQL percent defective*	*Minimum sample size*
Component material testing	L2	*	3

* Acceptance number shall be zero (0) for all sample sizes.

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4.2.3 *For chemical analysis of the metals.* Four ounces of each lot of each metal used in the construction of the cases shall be selected for chemical analysis as specified in 4.4.1.

4.2.4 *Component materials.* Sufficient samples of the component materials shall be selected from each lot of material used in the manufacture of the cases to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for component materials covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated in 4.2.2. The unit of product for testing shall be the quantity necessary to perform all required tests one time each, taking into consideration the fact that, where possible, the same material may be used to perform two or more tests.

4.3 Examination. The cases shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Chemical analysis of the metals.* The samples selected in accordance with 4.2 shall be analyzed in conformance with method 111 or 112 of Test Method Standard No. 151. In case of dispute, method 111 shall be used for referee analysis.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.* One (1) case, as specified, constitutes one unit.

5.1.2 *Packaging quantities.* The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I. *Packaging quantities*

Unit package	*Intermediate package*	*Exterior container*
1 unit	6 units	72 units

5.1.2.1 *Packing variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.* Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 *Intermediate package.* Intermediate package shall be a carton of appropriate size constructed in accordance with Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level C.* Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* Exterior container shall be designed for type 2 load, and constructed in accordance with Specification PPP–B–585, style 3 for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or MIL–B–10377, table II.

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5.2.1.2 *Case liner.* Each level A box shall be lined with a waterproof case liner conforming to Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.* Exterior container shall be constructed in accordance with Specification PPP–B–621, table I, or Specification PPP–B–636, type 1, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Interior packages.* Each unit package and each intermediate package shall be marked in accordance with section 5 of Standard MIL–STD–129.

5.3.2 *Exterior container.* Exterior container shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** This case is used to contain, protect, and hold in place an assemblage of dental instruments to form a subassembly of Stock No. 6545–927–4840, Dental Instrument and Supply Set, Emergency Treatment, Field.

6.2 **Availability of color chips.** The individual 3- by 5-inch chips of the colors referenced in 3.2.3 are available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S.W., Washington 25, D. C. Price 5 cents.

6.3 **Ordering data.** Procurement documents ments should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.5 **Transportation description.** Transportation description applicable to this item is:

Cases, Carrying, NOIBN, in boxes.

Carload minimum weight 15,000 pounds subject to Rule 34 of the Uniform Freight Classification.

Truckload minimum weight 5,000 pounds subject to Rule 34 of National Motor Freight Classification.

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